

AI Kitchen: Child-Readable Ingredient Labels and Transparency Stickers for Failure Anticipation and Reliance Calibration

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Abstract

Children increasingly meet AI inside creative tools, yet may treat model behavior as magic and read confidence as correctness. We built **AI Kitchen**, a web-based make-test-share experience that turns a tiny training set into **ingredients** and makes coverage gaps visible at decision time. An **Ingredient Label** summarizes the selected recipe, shows coverage bars for conditions (e.g., low light, motion blur, look-alikes), and generates deterministic, child-readable **predicted-failure hints** from missing coverage. During testing, a **predict-before-reveal** step asks children to guess whether the model will be right, wrong, or uncertain; a **Confidence Trap** then presents controlled high-confidence-wrong and low-confidence-right cases and asks children to **trust**, **verify**, or **retrain**. When sharing an artifact, the system exports a compact **Transparency Sticker** that travels with the artifact and highlights blind spots plus the reminder High confidence $\neq$ correct. We report the implemented prototype and instrumentation, and outline a **PLANNED** formative study (target N=10, ages 8–12) using log-based measures of failure anticipation, transparency use, and reliance decisions.

CCS Concepts

• **Human-centered computing** → **User interface design**; • **Applied computing** → **Education**.

Keywords

children, AI literacy, transparency, trust calibration, creative coding, model documentation

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1 Introduction

AI features (e.g., image classifiers, smart suggestions) are now common in youth-facing creative tools. Prior work on children’s AI literacy notes that learners need support to reason about what data a model learns from and when it may fail, rather than defaulting to agentic explanations or hindsight-only interpretations [3, 6]. This matters in practice: children are often asked to act on model outputs, and a confidence display can be misread as a guarantee.

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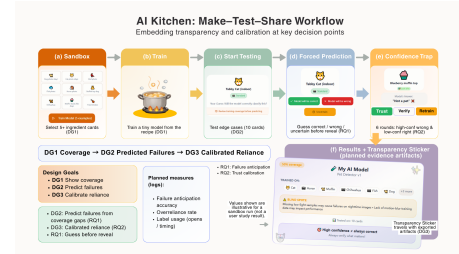


Figure 1: AI Kitchen workflow overview. Children select ingredient cards to form a tiny training recipe (DG1), test with predict-before-reveal and coverage-grounded failure hints (DG2), practice reliance decisions in a Confidence Trap (DG3), and export a Transparency Sticker that travels with shared artifacts (RQ1–RQ2).

This poster treats algorithmic literacy as an **interaction design problem**: how can an interface help children form *testable expectations* about failures and make *reliance* actionable inside a make-test-share workflow? We explore two questions:

- **RQ1: Failure anticipation.** Can a child-readable representation of training data and coverage help children predict when a model will fail?
- **RQ2: Reliance calibration.** Can short micro-interactions help children distinguish confidence from correctness and choose when to verify or retrain?

The **cooking metaphor** is not merely decorative: it maps onto key properties of machine learning that are otherwise abstract for children aged 8–12. Unlike a *map* (which implies a static, look-up relationship) or a *toolbox* (which implies interchangeable, reversible selection), cooking is a **transformative and irreversible process**—you cannot un-mix batter, just as you cannot separate training examples back out of a trained model. This irreversibility makes the core intuition concrete: what you put in *permanently shapes* what the model can recognize, so leaving out night-time images is like forgetting an ingredient—the dish will always be missing that flavor. Research on concrete operational thinking further supports food-based schemas as accessible to this age group, where tangible cause-and-effect chains (add → change) are more readily reasoned about than abstract set-membership [6].

Accordingly, this submission’s primary contribution is an **interaction design and conceptual contribution**: a set of novel, workflow-timed transparency mechanisms and child-readable artifacts whose design rationale can be evaluated and built upon independent of a completed empirical study.

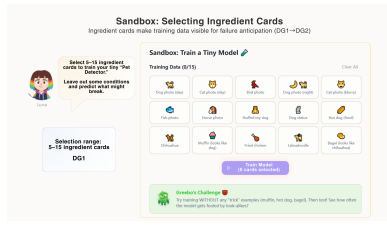


Figure 2: Sandbox ingredient selection (DG1). Training examples are represented as ingredient cards with simple condition tags. The selected set becomes the recipe summarized later by the Ingredient Label.

2 Background (brief)

Children’s AI literacy work argues for competencies around data, capabilities, and limitations, and shows that hands-on training/coding can shift perceptions while leaving persistent misconceptions about how models generalize [3, 6]. Family contexts can further shape how children interpret and talk about AI systems [2]. Separately, making-oriented platforms such as Scratch normalize iteration and sharing, suggesting a natural place to embed lightweight reflection into creation workflows [8].

In responsible AI, model cards and datasheets propose structured documentation of intended use and limitations [4, 7]. Human-AI interaction guidance and trust literature emphasize making system boundaries legible and supporting *appropriate reliance* rather than maximizing trust [1, 5]. AI Kitchen adapts these ideas into child-readable, workflow-timed artifacts: a coverage-first Ingredient Label used during testing, and an exportable Transparency Sticker for sharing.

3 System and interaction design

3.1 Design goals and prototype scope

AI Kitchen is a functional web prototype for ages 8–12. It is not a curriculum; the contribution is a small set of **interaction mechanisms** that make data coverage and reliance decisions visible in-the-moment:

- **DG1 Make data visible:** training ingredients and counts are legible.
- **DG2 Make failure predictable:** coverage gaps become testable failure hypotheses.
- **DG3 Make reliance actionable:** children practice decisions (trust/verify/retrain) under conflicting cues. DG1–DG2 primarily support RQ1; DG3 supports RQ2.

3.2 Sandbox: ingredient cards (DG1)

Children build a tiny training recipe by selecting 5–15 ingredient cards. Each card is tagged with conditions (e.g., day/night, blurry, look-alike), enabling later coverage summaries and gap-based hints (Fig. 2).

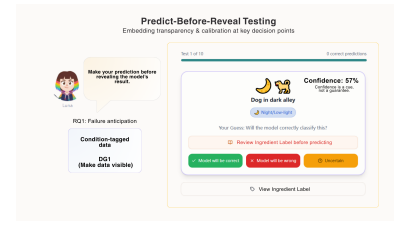


Figure 3: Predict-before-reveal (DG2). Each test card prompts a pre-reveal guess (correct/wrong/uncertain). The UI can nudge children to consult the Ingredient Label before committing to a prediction (RQ1).

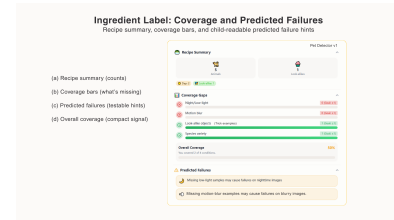


Figure 4: Ingredient Label (DG1–DG2). The drawer summarizes the recipe, visualizes coverage gaps, and turns missing coverage into child-readable predicted-failure hints intended to support test planning and failure anticipation (RQ1).

3.3 Predict-before-reveal testing (DG2; RQ1)

During testing, each trial requires a **forced prediction**—children guess whether the model will be correct, wrong, or uncertain before the answer is shown (Fig. 3). This makes failure anticipation observable and reduces hindsight-only explanations.

3.4 Ingredient Label: coverage and predicted-failure hints (DG1–DG2)

The **Ingredient Label** is a drawer that summarizes what the model saw and what it likely did *not* see. It includes (i) recipe counts, (ii) coverage bars for key conditions, and (iii) deterministic predicted-failure statements generated from missing coverage (Fig. 4).

These hints are generated by a **deterministic rule engine**: for each coverage dimension (e.g., lighting, motion, look-alike similarity), the system checks whether the child’s selected recipe contains at least one card tagged with that condition; if the count is zero, the corresponding failure hint is unconditionally activated and rendered in the label. For example, if no card tagged night or low-light is present in the recipe, the rule fires the hint “*I’ve never seen dark pictures—I might mess up at night.*” making the gap-to-prediction chain fully transparent and reproducible across participants. This links DG1 to DG2 and directly supports RQ1.

Three representative child-friendly hints produced by the rule engine are:

- “*I’ve never seen dark pictures—I might mess up at night.*”
- “*Blurry photos confuse me—I’ve only practiced with clear ones.*”
- “*Things that look really similar might trick me—I haven’t practiced telling them apart.*”

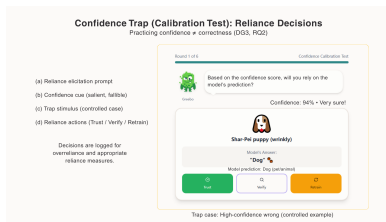


Figure 5: Confidence Trap (DG3). Each round shows a prediction plus a confidence cue and asks for an action (Trust/Verify/Retrain). Trap rounds deliberately decouple confidence from correctness to practice calibrated reliance (RQ2).

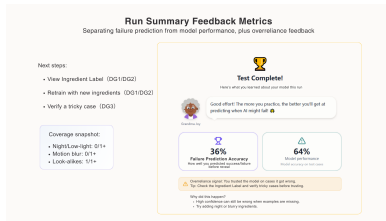


Figure 6: Run summary feedback. The results view distinguishes how well the child predicted failures from how well the model performed, and gives actionable next steps (review label, retrain, verify) when overreliance is detected.

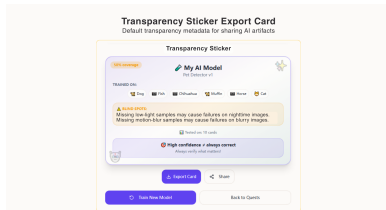


Figure 7: Transparency Sticker export. A shareable card summarizes the training recipe, flags blind spots derived from coverage gaps, and includes a reminder that confidence is not the same as correctness (DG3; RQ2).

3.5 Confidence Trap: reliance decisions with controlled cases (DG3; RQ2)

A six-round **Confidence Trap** presents controlled cases that include **high-confidence-wrong** and **low-confidence-right** outcomes. Children choose among **Trust**, **Verify**, and **Retrain** (Fig. 5), turning reliance into an explicit action rather than a vague attitude. The goal is practice: confidence is treated as a salient but fallible cue.

3.6 Run summary and transparent-by-default export (DG2–DG3)

After a run, the summary separates **failure-prediction accuracy** from **model performance** and surfaces an overreliance warning when children trusted wrong answers (Fig. 6). Export generates a compact **Transparency Sticker** that travels with shared artifacts and highlights the recipe and blind spots (Fig. 7).

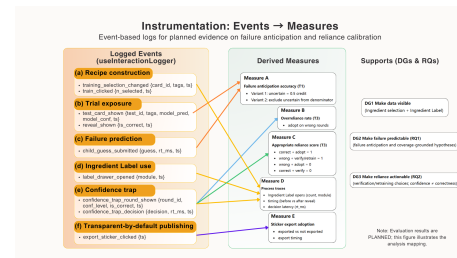


Figure 8: Instrumentation map (PLANNED analysis). Logged events (recipe construction, trial exposure, predictions, label use, reliance decisions, export) support derived measures for failure anticipation and calibrated reliance. No evaluation results are reported in this submission.

3.7 Technical truthfulness (scope control)

AI Kitchen is designed to study **interaction with transparency artifacts**, not to benchmark a classifier. The prototype uses **controlled/simulated outcomes** to reliably produce calibration scenarios (e.g., high-confidence-wrong rounds) and to generate deterministic failure hints from coverage gaps. Accordingly, any on-screen confidence should be interpreted as a **cue for practice**, not as a calibrated probability claim. This control lets us present repeatable calibration scenarios across participants.

4 Instrumentation (implemented) and planned measures

Every meaningful action in AI Kitchen generates a timestamped log event—recipe edits, label opens, per-trial predictions, Confidence Trap decisions, and export clicks. These traces feed two families of measures (Fig. 8): **failure anticipation** (did the child predict the right outcome, and did they consult the label first?) and **reliance quality** (did they trust, verify, or retrain—and was that the right call?). The instrumentation is already live in the prototype; the planned study will pipe events to JSON for analysis.

5 Planned formative study (PLANNED)

This submission reports a functional prototype and a **PLANNED** formative evaluation; **no human-subject results, quotes, or statistics are reported here**.

5.1 Participants and session flow (PLANNED)

We plan to recruit **N=10** children ages **8–12**—purposively sampled to include **diverse technical backgrounds and AI literacy levels** (e.g., children with and without prior coding experience, from both school and informal learning settings)—with parental consent and child assent, to capture a range of starting assumptions about model behavior. A single session (about 40 minutes) will: (1) brief introduction that the system can be wrong; (2) sandbox recipe selection and training; (3) predict-before-reveal testing (RQ1); (4) six-round Confidence Trap (RQ2); (5) run summary + export; (6) short post probes and a brief interview focused on how children explained failures and confidence.

5.2 Evidence criteria (PLANNED)

Given small N, we will report participant-level plots and robust summaries (median/IQR) and use conservative within-participant comparisons when appropriate (e.g., nonparametric tests). Mechanism-focused evidence would include: (i) improved failure anticipation on out-of-coverage conditions alongside label use at decision time; (ii) reduced overreliance (fewer Trust decisions on wrong high-confidence rounds); and (iii) more coverage-grounded explanations in talk, triangulated with logs (e.g., label opened before a correct model will be wrong prediction).

6 Design takeaways for child-facing transparency

From this prototype, we highlight three buildable patterns for child-facing AI transparency in creative tools:

- **Coverage-first transparency:** show what data the model did (and did not) see, and convert gaps into testable failure hints (Ingredient Label).
- **Predict-before-reveal:** elicit a commitment before feedback to make anticipation measurable and to reduce hindsight reasoning.
- **Transparency that travels:** make disclosure a default part of sharing via a compact, exportable sticker rather than a hidden about page.

7 Limitations

- **Scope of contribution.** This work is an **interaction design and conceptual contribution**—a functional prototype and design rationale for child-readable transparency mechanisms—rather than an empirical claim about learning outcomes. The planned formative study will test, but is not required to establish, the validity of the design patterns.
- Controlled/simulated outcomes support stable calibration scenarios but limit claims about real-world model calibration.
- Condition tags (e.g., low light, look-alike) simplify representativeness and may not capture richer dataset shifts.
- A short, single-session design does not test long-term retention or transfer to other tools.

Ethics and LLM Disclosure

This submission reports a functional prototype and a **PLANNED** study with children; the study will proceed only after appropriate ethics/IRB approval, with parental consent and child assent. An LLM was used only for editing; it was not used to generate empirical results (none are reported) or to invent sources, and citations were checked by the authors.

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